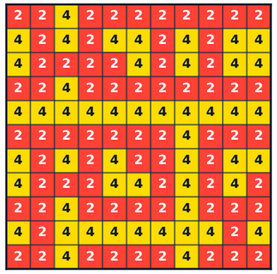
Figure 6 | Unified Differentiable Spectral-Green RKHS Operator: Forward & Implicit Adjoint Backward Pass

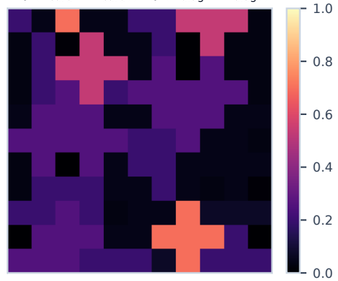
Trainable forward/backward normal-equation solver $\mathbf{G}_\lambda \mathbf{W}^* = \mathbf{\Psi}_S^\top (\mathbf{M}_S \odot \mathbf{Y}_S) + \boldsymbol{\lambda} \odot \mathbf{W}_0$ across spatial-topological equivalence classes

a Analytical Forward Pass: Learnable D_4-Steerable Wavelet Bank $\mathbf{\Psi}(X;\theta_\psi)$ & Woodbury-Green Linear Solve ($\mathbf{G}_\lambda \mathbf{W}^* = \mathbf{\Psi}_S^\top \mathbf{M}_S \mathbf{Y}_S + \boldsymbol{\lambda}\mathbf{W}_0$)

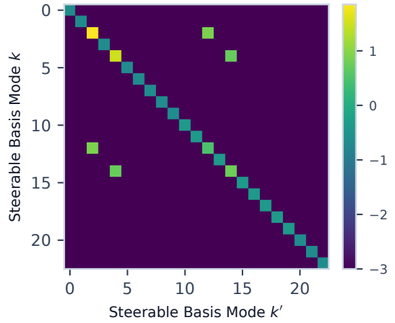
Query Patch X_q [8:19, 10:21]
Task 1818057f input lattice



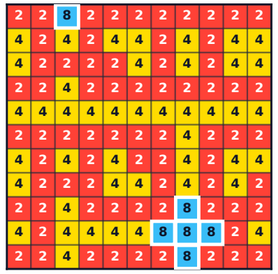
Learnable D_4 Wavelet Response $\chi_{D_4}(y,x)$
$\sigma(\beta(s_{\text{card}} - \tau_{\text{card}}))\sigma(\beta(d_{\text{diag}} - \tau_{\text{diag}}))$



Regularized Gram Matrix $\log_{10}|\mathbf{G}_\lambda|$
$\mathbf{G}_\lambda = \mathbf{\Psi}_S^\top \operatorname{diag}(\mathbf{m}_S)\mathbf{\Psi}_S + \operatorname{diag}(\boldsymbol{\lambda})$

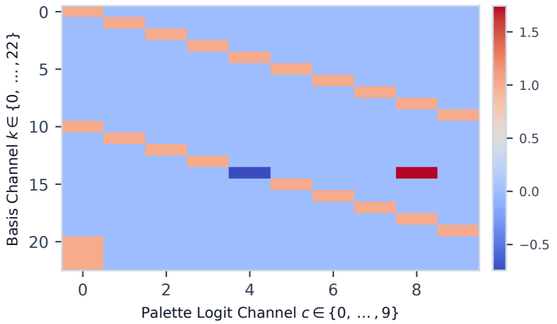


Forward Output $\mathbf{Z}_q = \mathbf{\Psi}_q \mathbf{W}^*$
100.0% Exact Match (0 Scaffold Leak)

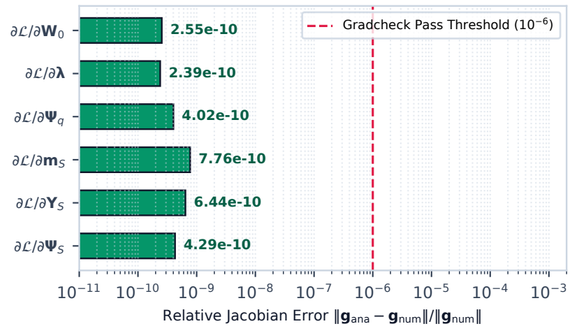


b Implicit Differentiation Through Regularized Normal Equations ($\mathbf{G}_\lambda \mathbf{\Lambda} = \mathbf{\Psi}_q^\top \tilde{\mathbf{Z}}_q$) & FP64 Gradcheck (Max Error $7.76e-10$)

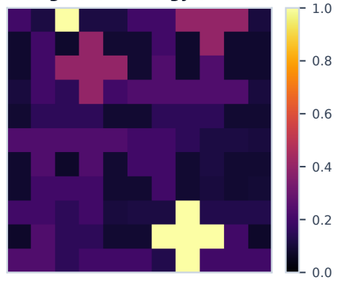
Optimal Operator Weights $\mathbf{W}^* = \mathbf{G}_\lambda^{-1}\mathbf{R}$
Solved in closed form without unrolling



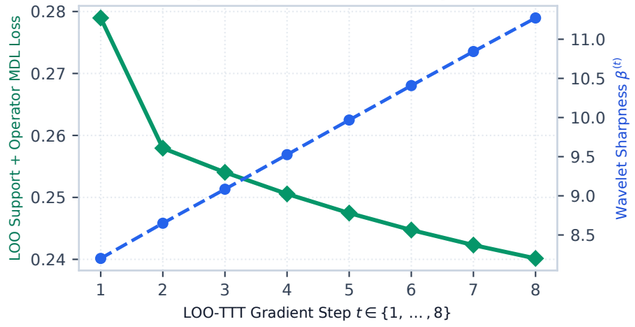
Full torch.autograd.gradcheck (FP64)
All 6 analytical adjoints exact ($\leq 7.76e-10$)



Adjoint Spatial Sensitivity $\|\partial\mathcal{L}/\partial\mathbf{\Psi}_q\|_2$
Focuses gradient energy on cross loci



Live 8-Step Operator Adaptation
Adjoint optimization of θ_ψ & $\log\boldsymbol{\lambda}$

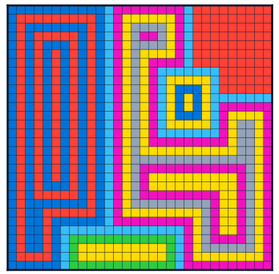


c Generalization of the Differentiable Operator Suite Across Spatial-Topological Equivalence Classes (0 TTT Steps)

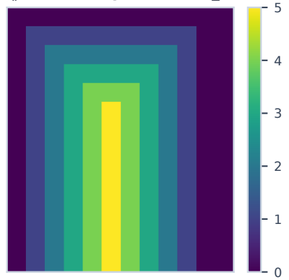
Task 0934a4d8 Solved (9x3)
100.0% Exact (27/27 px, Cavity Basis)



Task 13e47133 Eikonal (30x30)
100.0% Eikonal (900/900) | 99.44% 0-TTT Full



Model Chebyshev Eikonal Tensor d_∞
$\|\nabla_\infty d_\partial\| = 1$ **in `output.eikonal_state`**



Learnable Gaussian-Bessel Harmonics
Trainable calibers c_k & bandwidths σ_k

